Test Exercise 2

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$y = X_1\beta_1 + X_2\beta_2 + \varepsilon = X_1 b_1 + X_2 b_2 + e$

OLS estimator of $\beta_1 \longrightarrow b_R = (X_1'X_1)^{-1}X_1'y$ (for restricted model)

$P = (X_1'X_1)^{-1}X_1'X_2$

(a) $E[b_R] = E\left[(X_1'X_1)^{-1}X_1'y\right] = (X_1'X_1)^{-1}X_1'\,E[y]$

Here, we wrote $E[X_1] = X_1$, because X_1 is non-random according to assumption A2.

$E[b_R] = (X_1'X_1)^{-1}X_1'\,E[X_1\beta_1 + X_2\beta_2 + \varepsilon]$

Likewise, assumption A_2 yields $E[X_1] = X_1$ and $E[X_2] = X_2$.

A6 states that β_1 and β_2 are fixed, so $E[\beta_1] = \beta_1$ and $E[\beta_2] = \beta_2$.

A_3 states that $E[\varepsilon] = 0$.

We can now write:

$E[b_R] = X_1^{-1}\underbrace{\cancel{X_1'^{-1}X_1'}}_{I}X_1\beta_1 + \underbrace{(X_1'X_1)^{-1}X_1'X_2}_{P}\beta_2 + 0 \iff$

$\iff E[b_R] = \underbrace{\cancel{X_1^{-1}X_1}}_{I}\beta_1 + P\beta_2 \iff$

$\iff \boxed{E[b_R] = \beta_1 + P\beta_2}$

(b) $var[b_R] = E\left[(b_R - E[b_R])(b_R - E[b_R])'\right]$

- $b_R - E[b_R] = (X_1'X_1)^{-1} X_1' y - [\beta_1 + P\beta_2] =$

$$= X_1^{-1} \underbrace{\cancel{X_1'^{-1} X_1'}}_{I} \cdot [X_1\beta_1 + X_2\beta_2 + \varepsilon] - \beta_1 - P\beta_2 =$$

$$\boxed{\xcancel{= \underbrace{X_1^{-1} X_1}_{I}\beta_1 + X_1^{-1} X_2\beta_2 + X_1^{-1}\varepsilon - \beta_1 - P\beta_2} \quad \xcancel{= \beta_1}}$$

$$= \underbrace{\cancel{X_1^{-1} X_1}}_{I}\beta_1 + \underbrace{(X_1'X_1)^{-1} X_1' X_2}_{P}\beta_2 + (X_1'X_1)^{-1} X_1'\varepsilon - \beta_1 - P\beta_2$$

$$= \cancel{\beta_1} + \cancel{P\beta_2} + (X_1'X_1)^{-1} X_1'\varepsilon - \cancel{\beta_1} - \cancel{P\beta_2} =$$

$$= (X_1'X_1)^{-1} X_1'\varepsilon$$

- Likewise, $(b_R - E[b_R])' = \left[(X_1'X_1)^{-1} X_1'\varepsilon\right]' = \varepsilon' X_1 (X_1'X_1)^{-1}$

We can now write:

$$var[b_R] = E\left[\left((X_1'X_1)^{-1} X_1'\varepsilon\right)\left(\varepsilon' X_1 (X_1'X_1)^{-1}\right)\right] \overset{\text{assumption A2}}{\Leftrightarrow}$$

$$\Leftrightarrow var[b_R] = (X_1'X_1)^{-1} X_1' \cdot E[\varepsilon\varepsilon'] \cdot X_1 (X_1'X_1)^{-1}$$

$E[\varepsilon\varepsilon'] = \sigma^2 I$, implied from A4 and A5, as was shown in Lecture 2.4.1

So $var[b_R] = X_1^{-1} \underbrace{\cancel{X_1'^{-1} X_1'}}_{I} \sigma^2 I \underbrace{\cancel{X_1 X_1^{-1}}}_{I} X_1'^{-1} \Leftrightarrow var[b_R] = \sigma^2 X_1^{-1} X_1'^{-1} \Leftrightarrow$

$$\Leftrightarrow \boxed{var[b_R] = \sigma^2 (X_1'X_1)^{-1}}$$

(c) $b_R = (X_1'X_1)^{-1}X_1'y$

Substituting $y = X_1b_1 + X_2b_2 + e$, we get:

$$b_R = X_1^{-1}\cancel{X_1'^{-1}}X_1'X_1b_1 + \underbrace{(X_1'X_1)^{-1}X_1'X_2}_{P}b_2 + (X_1'X_1)^{-1}X_1'e \Leftrightarrow$$

$$\Leftrightarrow b_R = b_1 + Pb_2 + (X_1'X_1)^{-1}X_1'e$$

Since e is the residual vector from the unrestricted OLS regression of y on $X = [X_1\ X_2]$, the OLS equations imply:

$$\boxed{X'e = 0}$$

Because X_1 is a subset of the columns of X, this also implies:

$$\boxed{X_1'e = 0}$$

We can now write:

$$\boxed{b_R = b_1 + Pb_2}$$

(d) $P = (X_1'X_1)^{-1} X_1' X_2$

If we denote X_2 as Y, P can now be written as:

$P = (X_1'X_1)^{-1} X_1' Y$, which points to the OLS estimator.

However, $X_2 = [X_{Age} \; X_{Educ} \; X_{Parttime}]$, with X_{Age}, X_{Educ}, and $X_{Parttime}$ the (500×1) vectors with the observations of the variables 'Age', 'Educ', and 'Parttime' respectively.

Also, $X_1 = [X_{const} \; X_{Female}]$, with X_{const} and X_{Female} the (500×1) vectors for the constant term and the variable 'Female'.

Then P can be written as:

$P = [(X_1'X_1)^{-1} X_1' X_{Age}, (X_1'X_1)^{-1} X_1' X_{Educ}, (X_1'X_1)^{-1} X_1' X_{Parttime}]$, from which we can argue that each column of the matrix P is the OLS coefficient vector obtained by regressing 'Age', 'Educ', and 'Parttime' respectively on the regressors 'constant' and 'Female' in X_1.

(e) Revisiting Lecture 2.1, the regressions of each of the variables 'Age', 'Educ', 'Parttime' gave:

$$Age = 40.05 - 0.11\,Female + e$$

$$Educ = 2.26 - 0.49\,Female + e$$

$$Parttime = 0.20 + 0.25\,Female + e$$

According to the argument of question (d), P can be expressed:

$$\underset{(2\times3)}{P} = \begin{bmatrix} 40.05 & 2.26 & 0.20 \\ -0.11 & -0.49 & 0.25 \end{bmatrix}$$

(f) The values of b_R obtained by regressing Log-Wage on $X_1 = [\text{'constant'}, \text{'Female'}]$ are:

$$b_R = \begin{bmatrix} 4.73 \\ -0.25 \end{bmatrix} \quad \text{(from Lecture 2.1)}$$

Checking validity of result in question (c) $\rightarrow$ $\boxed{b_R = b_1 + P b_2}$:

$$\underset{(500\times1)}{Y} = \underset{(500\times2)}{X_1}\,\underset{(2\times1)}{b_1} + \underset{(500\times3)}{X_2}\,\underset{(3\times1)}{b_2} + \underset{(500\times1)}{e}, \quad b_1 = \begin{bmatrix} b_{constant} \\ b_{Female} \end{bmatrix} \quad b_2 = \begin{bmatrix} b_{Age} \\ b_{Educ} \\ b_{Parttime} \end{bmatrix}$$

Substituting the results given in Lecture 2.5 regarding the b values:

$$b_1 = \begin{bmatrix} 3.053 \\ -0.041 \end{bmatrix} \qquad b_2 = \begin{bmatrix} 0.031 \\ 0.233 \\ -0.365 \end{bmatrix}$$

We now check:

$$b_1 + P\,b_2 = \begin{bmatrix} 3.053 \\ -0.041 \end{bmatrix} + \begin{bmatrix} 40.05 & 2.26 & 0.20 \\ -0.11 & -0.49 & 0.25 \end{bmatrix} \cdot \begin{bmatrix} 0.031 \\ 0.233 \\ -0.365 \end{bmatrix} =$$

$$= \begin{bmatrix} 1.242 + 0.527 - 0.073 + 3.053 \\ -0.0034 - 0.114 - 0.09 - 0.04 \end{bmatrix} = \begin{bmatrix} 4.75 \\ -0.25 \end{bmatrix}$$

So, $b_R = \begin{bmatrix} 4.73 \\ -0.25 \end{bmatrix}$ and $b_1 + P\,b_2 = \begin{bmatrix} 4.75 \\ -0.25 \end{bmatrix}$, so the result in part (c) is valid.